Research Memo: Injector Turn-Off / Recirculation Circuit

Task: Close the gap `injector_boost.cir`'s RESULT NOTE names — turn-off is not modelled, no clamp/recirculation path is drawn anywhere in the design, and interrupting 18 A in 200 µH releases ~32 mJ per event with nowhere defined for it to go. This memo establishes what real automotive injector drivers do about that, before a SPICE block gets built. **Date:** 2026-09-18 **Scope:** Read-only research. No repo files modified, nothing committed.

Verdict (read this first)

Production peak-and-hold injector drivers do **not** use a plain freewheel diode for turn-off — that topology is real (it's the textbook baseline and the cheapest option) but it is also the slowest, and slow turn-off is exactly what this project's boost rail was built to avoid on the *opening* side. The turn-off side has the same problem in reverse. Two topologies dominate production practice for this injector class: **active/Zener clamping** (fast, dissipative, tunable voltage) and **recirculation of the coil's turn-off energy back into the boost reservoir capacitor** (fast, largely non-dissipative, and confirmed as real automotive practice going back to at least a 1988 Magneti Marelli diesel-injector patent, not a textbook idealization). For this design, **recirculation to the existing 100 V / 47 µF boost rail is the recommended topology**, clamped by a diode rated for the boost-rail voltage: it collapses 18 A in 200 µH in about **36 µs** (physics below), and the ~32 mJ recovered per event is the same order of magnitude as the ~7.3 V peak-phase droop `boost_converter.cir` already calculated — this is not a rounding error, it changes that block's charge balance.

Sourcing note, stated as plainly as memo 09 states its own: this pass retrieved one strong, directly on-topic primary source in full (Nexperia AN50003, an application note that surveys exactly the four candidate topologies the task asked about, with a quantified loss/performance comparison table) and one confirming patent (Marelli Autronica, 1988, Google Patents full text) that is diesel-injector-specific and explicitly diesel-electro-injector. Several other leads that looked promising by title (ST L9781, NXP AN4849, the ISO 7637-2 standard text itself) **could not be retrieved** — repeated 403s, timeouts, or a 404 — and are recorded as negative results below rather than filled in from memory or a search snippet dressed up as a document. TI DRV8701 was checked and ruled

out: it is a brushed-DC-motor H-bridge driver, not an injector-class part, despite surface-level keyword overlap ("solenoid control" appears in its marketing copy).

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1. What actually absorbs the coil energy at turn-off

1.1 Nexperia AN50003, "Driving solenoids in automotive applications" — CONFIRMED, retrieved and read in full

https://www.digikey.com/Site/Global/Layouts/DownloadPdf.ashx?pdfUrl=7547AD78C870436881732FA36D2AAAFA (Nexperia B.V., Rev 1.0, 4 December 2020 — an automotive-solenoid-driver application note, explicitly citing "peak-and-hold" fuel injection current profiles as its motivating example, §3, Fig. 3)

This is the single best-matched source found in this pass: it surveys **exactly** the four candidate topologies named in the task, with SPICE-based loss and performance numbers (their reference case: 12 V battery, 5 mH inductor, 3 A peak, 1.2 A hold — a much smaller solenoid than this injector, but the physics and the relative comparison transfer directly). Quoting Table 1 and Table 2 (§5) in full:

Topology	Total energy loss (mJ)	Cost	Speed	Efficiency	Reliability
Free-wheeling diode	17.0	Low	Low	High	Long term
MOSFET avalanche	20.7	Low	Medium	Low	Long term
Active clamp (Zener, drain-to-gate)	20.7	Low	Medium	Low	**Questionable**
Boost converter (energy recovery)	26.0	High	High	High	Long term

Their own text on the "Questionable" reliability rating for active clamp (§4.3), quoted directly: *"During active clamp there are high energy charge carriers generated in proximity of the MOSFET's gate oxide. These carriers might be injected into the oxide and cause damage. Over many active clamp cycles the gate oxide can wear out and cause parametric shift and ultimately device failure. Currently it is not recommended to use MOSFETs in repetitive active clamp."**This directly rules out the naive Zener-across-the-FET version of "active clamp" for a component that will do this thousands of times a minute for the life of the genset.**

On the boost/recovery topology (§4.4), quoted: *"During the Peak and Hold Phases the nominal battery voltage is switched, as in the previous cases. This allows for fast actuation of the solenoid, but also the energy from the solenoid is regenerated into the DC link capacitor of the boost converter... Although the losses appear to be higher in the Boost topology, the recuperation of the energy means that its efficiency is on par with the free-wheeling topology, despite using higher R_DSon*

components." (§5) — i.e. the boost/recovery topology has the **highest absolute component losses in their table but the best net efficiency**, because most of the 32 mJ-scale energy is returned to the rail rather than burned.

On the free-wheeling diode being the slowest (§4.1), quoted: *"Compared to the other driver topologies, the free-wheeling driver is simple, has a low component count, but it is the slowest due to the inductor voltage being approximately equal to the battery voltage."*

1.2 The industry-standard schematic vocabulary — CONFIRMED across two independent IC datasheets

- **Infineon TLE8242-2** datasheet, retrieved in full (rev 1.0, Feb 2010): https://www.infineon.com/dgdl/Infineon-TLE8242_2-DS-v01_00-en.pdf — this is a **low-side constant-current pre-driver IC**, explicitly stated to be for "Variable Force Solenoids (e.g. automotive transmission solenoids)" and other constant-current solenoids (EGR, idle air control), not a diesel common-rail injector driver — **important scale mismatch**: its current range is 0–1.2 A and its absolute max POSx/NEGx rating is 50 V, roughly an order of magnitude below this injector's 18 A / 100 V. It is cited here only for what it confirms about topology convention, not for numbers. §1.3.1 and §1.3.2, quoted directly: *"Note: An external flyback clamp is required in [Direct PWM] configuration otherwise the IC may be damaged"* and *"Note: An external recirculation diode is required in [constant current mode] configuration otherwise the IC may be damaged."* **The IC itself never integrates the clamp/recirculation element — it is always an external, designer-chosen part**, which matches what this project needs to design rather than buy.
- **ST L9781 — SNIPPET only, not retrieved.** Two direct fetch attempts (st.com PDF, alldatasheet mirror) both failed — one timed out, one returned HTTP 403. What search-result snippets say (unverified against the document itself): the L9781 is a multi-valve pre-driver with an integrated boost step-up converter reported as "V-boost up to 80 V," and its recirculation/clamp behaviour is described in a snippet as *"Fast recirculation happens through clamp activation, with S2 open and clamp on S1 activated"* — language consistent with recirculation back toward the boost-side switch/rail rather than a simple flyback-to-ground diode. This is recorded as a data point corroborating the recirculation-to-boost topology, but it is explicitly **not** elevated to CONFIRMED status because the datasheet itself was never read.

1.3 Energy recovery in real diesel-injector drive circuits — CONFIRMED via patent full text (Google Patents), not a datasheet

US Patent 4,862,866, "Circuit for the piloting of inductive loads, particularly for operating the electro-injectors of a diesel-cycle internal combustion engine," assignee **Marelli Autronica S.p.A.** (Magneti

Marelli's electronics arm — a genuine Tier-1 diesel-ECU supplier, a Bosch competitor), filed 22 August 1988. https://patents.google.com/patent/US4862866A/en

Quoted directly from the retrieved claims/description text: the circuit includes *"energy-recovery circuit means controlled by the unit and adapted to enable part of the reactive energy stored in the load to be recycled towards the supply VB each time a load is deactivated."* This is explicit, diesel-electro-injector-specific, decades-old confirmation that **energy recovery at injector turn-off is real production-line thinking, not a textbook flourish invented for this memo.** No numeric clamp voltage or recovered-energy figure is given in the retrieved text — the patent describes the mechanism (an OR-diode network into a junction with a recovery switch SW4 back to the supply), not performance numbers.

A second patent, EP0622536A2 (Chrysler, priority 1993), was also retrieved via Google Patents and is recorded for completeness but is **not** treated as strong evidence for this design: it is a **gasoline two-stroke direct-injection** patent, not diesel common-rail, and its numbers (flyback clamped to "15–20× battery potential," i.e. ~180–240 V on a 12 V system, against an 8–10× battery boost voltage of ~96–120 V) describe a self-boosted clamp stage deliberately driven *above* the boost rail, a different design choice than recirculating into the existing boost capacitor. Noted as related prior art, not as a number to adopt.

1.4 Bosch EDC17/EDC7 architecture — negative result

No public technical description of Bosch's own injector-driver ASIC internals (clamp topology, voltage, or recirculation) was found. Search results for EDC17 return tuning/remapping community material (map structures, torque-based control philosophy) with no circuit-level disclosure. This is consistent with memo 04's own finding that OEM injector-driver silicon and calibration data are not publicly documented — recorded here as a negative result, not filled in by inference from the adjacent (and more open) NXP/Nexperia/Infineon material.

1.5 TI DRV8701-class — ruled out, not applicable

The DRV8701 datasheet (https://www.ti.com/lit/ds/symlink/drv8701.pdf) is a single H-bridge **gate driver for a 5.9–45 V, bidirectional brushed-DC motor**. It is not a peak-and-hold constant-current injector driver and has no boost-voltage handling in this part's role. It surfaced in search results only because "solenoid control" appears in TI's generic marketing description for the DRV87xx/DRV870x family. **Recorded as a checked-and- rejected candidate, not a silent omission.**

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2. Clamp voltage vs turn-off time — the actual numbers

2.1 The physics, cross-checked against a retrieved source

`di/dt = V_clamp / L` while the coil discharges above zero, so `Δt ≈ L·ΔI / V_clamp`. This is not a new derivation — it is exactly the relation Nexperia's AN50003 uses (§4.1, Eq. 1: `Δt = L·ΔI/V`), and their own worked numbers cross-check cleanly: their free-wheeling case (12 V, 5 mH, 1.2 A hold current) gives `Δt = 5e-3 × 1.2 / 12 = 0.5 ms`, matching their stated "close to 0.5 ms" (§4.1). Their avalanche case, clamped at a measured 68 V (Fig. 7, read directly off their MOSFET avalanche plot) against the same 5 mH/1.2 A step, gives `Δt = 5e-3 × 1.2 / 68 ≈ 88 µs`, and their text states the reduction is "close to... a five-fold reduction" — 12/68 ≈ 0.176, i.e. a 5.7× voltage ratio, consistent with their "five times faster with five times larger voltage" framing (§4.2). **This cross-check is original arithmetic done for this memo against Nexperia's own published numbers, not itself sourced — marked INFERRED, but it validates that the formula and their reported figures agree with each other.**

2.2 Applied to this injector — INFERRED, derived, not sourced for this specific part

Using this project's own numbers (18 A, 200 µH, from `injector_boost.cir`):

Clamp path	Clamp voltage	Turn-off time (18 A → 0)	Note
Freewheel to battery	~13.5–14 V	**~257–267 µs**	Baseline, slowest
Avalanche/Zener active clamp	68 V (Nexperia's example FET rating)	**~53 µs**	~5× faster, matches their ratio
Recirculate to boost rail	100 V (this project's existing rail)	**~36 µs**	Recommended path, see §3
Dedicated higher clamp above boost	150 V (illustrative)	**~24 µs**	Faster still, needs a part rated well above the 100 V rail

At a 1–3 ms injection duration (memo 04's envelope, corroborated by `injector_boost.cir`'s own framing), even the slowest option (267 µs, battery freewheel) is a smaller fraction of the injection window than the 38 µs turn-on ramp is — but turn-off precision matters disproportionately for **small-quantity pilot injections**, where end-of-delivery timing directly sets fuel quantity. No source in this pass gave a quantified "needle closes N µs after current reaches zero" mechanical lag figure for this injector class — that remains an open item, same as memo 04's Step 5 calibration gap. **This whole table is INFERRED (derived arithmetic), not a number pulled from a source for this specific injector** — the only source-backed clamp-voltage figure found in this pass is Nexperia's own 68 V MOSFET avalanche example, which is a generic automotive part rating, not an injector-specific spec.

2.3 Why higher clamp voltage is not free — confirmed qualitatively, not quantitatively

AN50003 states plainly (§4.2, §4.3) that faster (higher-voltage) turn-off concentrates the *same total energy* into a shorter, hotter pulse, and that this is the mechanism behind their "Questionable" reliability flag on active clamp (gate-oxide stress) and their explicit statement that existing MOSFET datasheets give only "scarce data" on repetitive avalanche ratings (§6). **No source in this pass gave a quantitative die-temperature-rise-per-volt figure** — that would require a specific FET's thermal datasheet and is a detailed-design question, not a topology question. Recorded as an open item rather than estimated.

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3. Energy recovery — real practice or textbook complication

Real practice, with a caveat about how much it matters here.

- Confirmed via the Marelli Autronica 1988 patent (§1.3 above): energy recovery at diesel-injector deactivation is documented production-supplier engineering, not a modern textbook idea being retrofitted onto this project.
- Confirmed via Nexperia AN50003 (§1.1 above): the boost-recovery topology has the best net efficiency of the four surveyed, specifically *because* turn-off energy is regenerated into the DC-link capacitor rather than burned.
- SNIPPET-level (ST L9781, not retrieved in full) is consistent with the same idea in a currently-sold automotive injector pre-driver part.

Does it meaningfully change this project's boost converter load? This is where the sourcing runs out and the memo has to say so: **no source found gives a quantitative "% duty cycle reduction" figure for recovery vs. no recovery on a diesel common-rail boost stage.** What follows is original arithmetic against this project's own `boost_converter.cir` numbers, marked **INFERRED**:

- Coil turn-off energy per event: ~32.4 mJ (`0.5 × 200µH × 18A² = 32.4 mJ`, matching `injector_boost.cir`'s RESULT NOTE figure of "about 32 mJ").
- `boost_converter.cir`'s reservoir is 47 µF at a 100 V setpoint, and its own RESULT NOTE calculates a **7.3 V droop** from the peak-phase charge draw (342 µC / 47 µF).
- If that same 32.4 mJ were returned to the same 47 µF/100 V capacitor (energy balance `ΔE = C·V·ΔV`, linearised around 100 V): `ΔV = ΔE / (C·V) = 32.4e-3 / (47e-6 × 100) ≈ 6.9 V`.
- **That recovered-voltage bump (≈6.9 V) is essentially the same size as the droop the peak-phase draw already causes (7.3 V).** This is not a small correction — on this project's own reservoir sizing, full recovery

would come close to cancelling the very droop `boost_converter.cir` exists to characterize. It would change the charger's required average current (`Bchga`'s 50 mA figure) and could push the rail *above* its 100 V setpoint on back-to-back events if the regulator doesn't shed the excess, which is exactly the "climbed past target, reported 122 V" failure mode that block's own header describes for an unregulated charger.

This is the concrete reason recovery is worth building here, not a default recommendation copied from the literature: it isn't a minor efficiency nicety, it is the same order of magnitude as a number this project has already calculated and cared about.

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4. Thermal — events per second and average dissipation

Assumptions stated plainly, because the task brief itself contains a contradiction worth flagging: the task text refers to "a 4-cylinder at 1500 rpm genset speed" in one clause and then names the actual engine as "3GK550ETA 4SR1, 3-cylinder" in the same sentence. Per memo 04 (confirmed, engine identity task), **this engine is a 3-cylinder** (Kirloskar 3GK550ETA 4SR1, 3-cyl inline, KG4-25WS1 genset). This memo uses **3-cylinder**, not 4, and flags the discrepancy rather than silently picking one.

Derivation (original calculation, INFERRED, following memo 04's own Step 3 method exactly): - At 1500 rpm: one crank revolution = 40 ms; a 4-stroke cycle (720°) = 2 revolutions = **80 ms** (matches memo 04 and `boost_converter.cir`'s own 26.67 ms per-cylinder figure: 80/3 = 26.67 ms). - Assumption: pilot + main + post injection per cylinder per cycle = **3 injection events per cylinder per 720° cycle**. This is stated as an assumption because no source in this pass (or in memo 04) gives a confirmed injection-strategy count for this specific engine/calibration — it is a reasonable assumption for a CPCB IV+-class common-rail genset (memo 04 already assumes pilot-injection capability is a hard requirement for its minimum-pulse-width figure), not a measured fact. - Total events per cycle: 3 cylinders × 3 injections = **9 events / 80 ms**. - **Events per second: 9 / 0.080 s = 112.5 Hz.**

Worst-case average dissipation, assuming every one of the 9 events collapses from the full 18 A peak (i.e., assuming a purely dissipative clamp — freewheel-to-battery or avalanche/Zener, not the recovery topology recommended in §3): `P_avg = 32.4 mJ × 112.5 /s ≈ 3.6 W`.

This is a **modest, not alarming, average number** — well within what a TO-220/DPAK-class clamp FET or Zener can shed continuously, even with a fairly restrained heatsink. The reason it still matters is not the average, it's the **peak instantaneous dissipation during each ~30–50 µs clamp interval** (32.4 mJ delivered in ~36–53 µs is a multi-hundred-watt

instantaneous pulse into the clamp element even though the time-averaged figure is small) — this is precisely the stress mode AN50003 flags as "Questionable" for active-clamp reliability (§1.1) and the reason its own avalanche-rated parts are only characterized up to single-digit-mJ per-event energies (their BUK9K35-60RA example: 4.9 mJ/event, ~2.5 billion cycle life at that energy, §6) — **an order of magnitude below this injector's 32.4 mJ per event.** No source in this pass characterizes a repetitive-avalanche or active-clamp part rated for 32 mJ/event at anything like injector-relevant cycle counts; this is recorded as an open item for the eventual FET selection, not answered here. **This is exactly the argument for recirculating the energy into the boost cap rather than dissipating it in a clamp device at all** — it sidesteps the question of whether a suitably-rated dissipative part even exists in this energy class.

If pilot/post events are assumed to turn off from hold current (~10 A) rather than full peak (plausible if they are short enough to be within the hold phase at cutoff, though memo 04 notes the boost ramp to 18A takes only ~38 µs, well under the ~100–300 µs minimum pulse width, so even short pilot pulses likely do reach peak current before cutoff) — the worst case above should be treated as the design figure, not a conservative upper bound to be discounted.

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5. ISO 7637-2 / CISPR 25 implications of the turn-off edge

Partially sourced — the standard's own text could not be retrieved.

- An attempt to fetch the ISO 7637-2:2011 standard text directly (a third-party PDF mirror) returned **HTTP 401 (paywalled/blocked)**. The official standard was not retrieved in this pass. What follows is drawn from search-result summaries describing the standard's pulse definitions — **SNIPPET level**, not confirmed against the standard itself.
- Per those snippets: **ISO 7637-2 pulse 2a** is specifically defined to simulate *"transients due to sudden interruption of currents in a device connected in parallel with the DUT due to the inductances of the wiring harness"* — this is, in general terms, structurally the same event class as an injector solenoid turn-off on a shared harness (an inductive load on the same supply/return network being suddenly interrupted). **This is an inference about applicability, not a confirmed statement from the standard that injector turn-off is pulse 2a's origin** — no source found states that explicitly for this project's harness.
- Pulses 3a/3b are described (snippet level) as driven by "distributed capacitance and inductance of the wiring harness" during switching processes — again structurally relevant to a fast clamped turn-off edge propagating down a wire that runs the length of the engine, but not confirmed against the standard text.

- `emi_filter.cir`'s own RESULT NOTE already establishes this project's relevant framing: it models **differential-mode conducted emissions only** (150 kHz–108 MHz, CISPR 25 Class 5, 18 dBµV average in the FM band per research memo 06), and explicitly defers common-mode analysis (which a fast dV/dt edge referenced to chassis would excite more than a differential edge) as "a specification, not a simulated result, until there is a layout to model it against."

What transfers, stated as INFERRED, not sourced: a clamped turn-off at 100 V in 36 µs is a dV/dt of roughly 100 V / 36 µs ≈ **2.8 V/µs** — two orders of magnitude gentler than the buck stage's ~400 kHz switching edges that `emi_filter.cir` already exists to filter (a 400 kHz square edge with even a few-hundred-ns risetime is tens to hundreds of V/µs). This suggests the turn-off edge is **unlikely to be the dominant conducted-emissions contributor** compared to the buck converter already being filtered, but this is a first-order comparison, not a simulated result, and it says nothing about the common-mode / radiated-emissions picture, which `emi_filter.cir` itself has not modelled for any source yet. **This is the one open item from this section worth carrying into the eventual SPICE block:** once a turn-off netlist exists, its edge should be checked against `emi_filter.cir`'s existing differential-mode model rather than assumed benign.

What transfers to this design

1. **Topology choice is a solved, four-way-classified problem** (Nexperia AN50003), not something to invent from first principles. Free-wheel is cheapest and slowest; avalanche and active-clamp are similar in speed and loss to each other but active-clamp carries a documented long-term reliability caution this project should not casually inherit; boost/recovery is the most complex and highest nominal component cost but the best net efficiency *and* is the one option with genuine diesel-injector production precedent (Marelli, 1988).
2. **The clamp-voltage-vs-speed trade-off is a straightforward L/V relation** that this memo's own arithmetic cross-checks cleanly against Nexperia's published numbers — no separate injector-specific figure was found or is needed to use the relation correctly.
3. **Recirculation to the boost rail is not a marginal optimization for this specific design** — the ~6.9 V recovered bump is the same order of magnitude as the ~7.3 V droop `boost_converter.cir` already treats as the central sizing question for that block. Building the turn-off circuit without considering recovery would leave that block's charge balance silently wrong once turn-off events are added to the picture.

4. **32.4 mJ/event is a real thermal design constraint for any dissipative clamp part**, even though the *average* power (3.6 W worst case) is modest — it is the peak/instantaneous stress during each 30–50 µs clamp event that matters, and no source in this pass identifies an avalanche-rated or Zener-clamp part characterized anywhere near this energy class per event. This is an argument in favor of recovery (minimizing energy actually dissipated in a switch/clamp device) rather than a solved problem to defer.
5. **The turn-off edge should be checked against, not assumed clear of,** the existing `emi_filter.cir` differential-mode model, and eventually against a common-mode model neither this memo nor that block has built.

Recommendation: the circuit to build

Topology: Low-side switch (per injector channel, matching the existing bank-shared high-side / independent low-side wiring memo 04 already verified) with a **recirculation diode from the injector's low side back into the 100 V boost rail capacitor** — the "boost converter" / energy-recovery variant from AN50003 §4.4, adapted from its PWM-ripple-recovery role (used there during the Peak/Hold chop) to also handle full End-of- Injection turn-off, which is the diesel-electro-injector-specific architecture the Marelli 1988 patent describes in general terms. This is **not** a Zener-across-the-FET active clamp (ruled out per §1.1's reliability finding) and **not** a bare freewheel-to-battery diode (too slow relative to the boost rail already built for the opposite reason on turn-on).

Clamp voltage: the existing **100 V boost rail itself** — no separate, higher clamp voltage is recommended. This keeps the clamp element's voltage rating aligned with parts already being selected for the boost stage, avoids introducing a fifth voltage domain, and gives a computed ~36 µs turn-off (§2.2), comfortably fast against the 1–3 ms injection window. A higher dedicated clamp (§2.2's 150 V illustrative row) would turn off faster still, but nothing in this pass's sourcing justifies paying for a second high-voltage rail when the existing one is already close to the useful range AN50003's own avalanche example (68 V) demonstrates.

Part class: a diode (ultrafast recovery, rated for boost-rail voltage and the 18 A peak current, matching the class TLE8242-2's own datasheet calls out generically — "recirculation diode (ultrafast)" — even though that IC itself is the wrong current/voltage class for this injector) in series with the existing low-side switch topology; no new active clamp IC is recommended. Should the boost rail's regulator (`Bchga` in `boost_converter.cir`) need to shed recovered energy that pushes it

above 100 V on back-to-back events, that is a boost-converter-block problem to solve in that block's next revision, not a reason to add a second dissipative clamp path here.

Specific claims the next SPICE block should check: 1. Turn-off time from 18 A to ~0 A when clamped to a 100 V rail: expect **~36 µs** (physics-derived, §2.2) — if the simulated result is far off this, the model's assumptions (diode drop, rail sag under load) need re-examination before trusting it. 2. Boost-rail voltage excursion from one full-energy recovery event on the existing 47 µF/100 V reservoir: expect **~+6.9 V** (§3) — compare against `boost_converter.cir`'s own ~7.3 V peak-phase droop to see whether recovery approximately cancels it, overshoots it, or undershoots it once diode/switch losses are included (this memo's number ignores those losses; AN50003's own data suggests recovery topologies still have real, non-zero losses even though net efficiency is favorable). 3. Whether recovered energy pushes the rail above the 100 V setpoint on back-to-back pilot+main+post events at the 26.67 ms per-cylinder spacing, and whether the existing `Bchga` charge-regulation logic needs a bleed path. 4. The clamp diode's peak instantaneous power/junction stress during the ~36 µs turn-off pulse, not just the ~3.6 W time-averaged figure (§4) — the average number is not the one that will size the part. 5. The turn-off edge's differential-mode contribution against `emi_filter.cir`'s existing model (§5) — first-order comparison here suggests it is gentler than the buck stage's edges, but this should be checked, not assumed.

Negative results, recorded

- **ST L9781 datasheet** — not retrieved. Two fetch attempts (st.com direct PDF, alldatasheet mirror) failed: one timed out (60 s), one returned HTTP 403. Everything attributed to it in this memo is explicitly marked SNIPPET and should not be used as a design number without retrieving the document.
- **NXP AN4849** ("Four Injector and Fuel Pump Drive Featuring the MC33816") — the PDF URL found via search (nxp.com/docs/en/application- note/AN4849.pdf) returned **HTTP 404**. Not retrieved. Search-result summaries describing its "boosting"/"holding" high-side driver pairing per bank are recorded in this memo's working notes but were not used as a cited finding above, since the document itself was never seen.
- **Bosch EDC17/EDC7 injector-driver ASIC architecture** — no public technical/circuit-level source found at all (see §1.4). This is a negative result about public availability, not evidence that Bosch's internal architecture differs from what's described here.

- **ISO 7637-2:2011 standard text** — one direct-PDF fetch attempt returned **HTTP 401.** Not retrieved. §5's discussion of pulses 2a/3a/3b is search-snippet level throughout and is flagged as such at each claim.
- **A quantitative die-temperature-rise-per-clamp-voltage figure** (task item 2's request for "actual numbers real drivers use," beyond the single 68 V generic-automotive avalanche example found) — not found. No injector-specific clamp-voltage number (as opposed to a generic solenoid or a 15–20×-battery gasoline-DI figure) was retrieved from any source in this pass. The 100 V recommendation in this memo's §"Recommendation" is a design choice justified by matching the existing boost rail, not a sourced industry-standard clamp voltage for this injector class.
- **A confirmed injection strategy (pilot/main/post count) for this specific engine/calibration** — not found; memo 04 does not establish this either. §4's "3 events per cylinder per cycle" is stated as an assumption, not a sourced fact.
- **TI DRV8701 relevance** — checked and explicitly ruled out (§1.5), not silently dropped, since the task named it as a candidate to check.